

General Chemistry

Chapter 8: Basic Concepts of Chemical Bonding

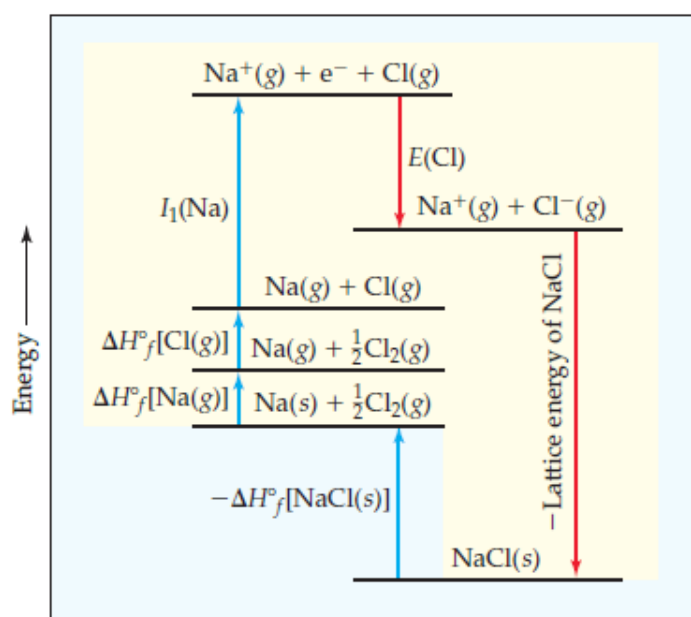
Review

Definitions:

- **General types of chemical bonds:** ionic bonds, covalent bonds and metallic bonds.
 - 1) Ionic bonds: electrostatic attraction
 - 2) Covalent bonds: electrons shared
 - 3) **Metallic bonds: electrons delocalized**
- **Valence electrons:** The electrons involved in chemical bonding, for most atoms are those in the outermost occupied shell.
- **Lewis electron-dot symbols (Lewis symbols):** consists of the element's chemical symbol plus a dot for each valence electron.
 - 1) The dots are placed on the four sides of the symbol—top, bottom, left, and right;
 - 2) Each side can accommodate up to two electrons;
 - 3) All four sides are equivalent;
 - 4) In general, spread out the dots as much as possible;
- **The Octet Rule:** Atoms tend to gain, lose, or share electrons until they are surrounded by eight valence electrons.

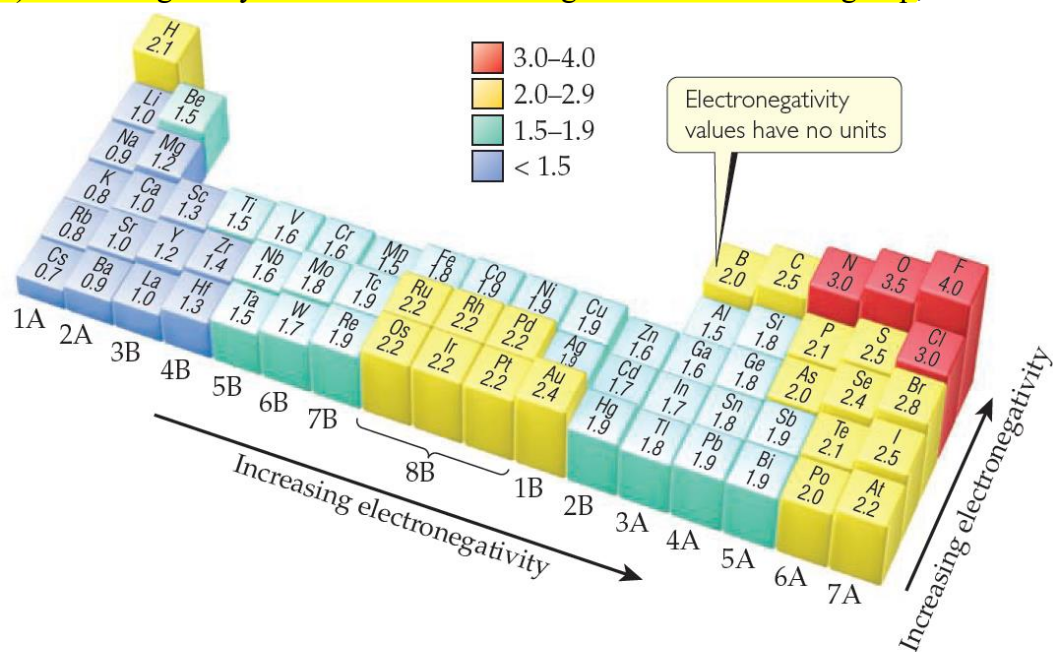


- **Lattice energy:** The energy required to completely separate one mole of a solid ionic compound into its gaseous ions.
 - 1) $\text{NaCl(s)} \rightarrow \text{Na}^+(\text{g}) + \text{Cl}^-(\text{g})$ $\Delta H_{\text{lattice}} = +788 \text{ KJ/mol}$
 - 2) **For a given arrangement of ions, the lattice energy increases as the charge on the ions increase and as their radii decrease.**



▲ **Figure 8.5 Born-Haber cycle for formation of NaCl.** This Hess's law representation shows the energetic relationships in the formation of the ionic solid from its elements.

- **Electron configuration of ions (s-,p-,d-block elements):**
 - 1) We expect ionic compounds of the representative metals from groups 1A, 2A, and 3A to contain 1+, 2+, and 3+ cations, respectively;
 - 2) We expect ionic compounds of the representative nonmetals from groups 5A, 6A, and 7A to contain 3-, 2-, and 1- anions, respectively;
 - 3) In forming ions, transition metals lose the valence-shell s electron first, then as many d electrons as required to reach the charge of the ion.
- **Covalent bond:** A chemical bond formed by sharing pairs of electrons.
- **Bonding pairs:** shared electron pairs
- **Lone pairs (nonbonding pairs):** unshared electron pairs
- **Single bond:** A shared electron pair
- **Double bond:** Two shared electron pairs between two atoms
- **Triple bond:** Three shared electron pairs between two atoms
 - 1) The bond length between two atoms decreases as the number of shared electron pairs increases.
 - 2) The bond enthalpy between two atoms increases as the number of shared electron pairs increases.
- **Bond polarity:** A measure of how equally or unequally the electrons in any covalent bond are shared.
- **Nonpolar covalent bond:** the electrons are shared equally.
- **Polar covalent bond:** the atoms exerts a greater attraction for the bonding electrons than the other.
- **Electronegativity:** the ability of an atom in a molecule to attract electrons to itself.
- **Trends:**
 - 1) Electronegativity increase from left to right across a period;
 - 2) Electronegativity decrease with increasing atomic number in a group;



▲ **Figure 8.7** Electronegativity values based on Pauling's thermochemical data.

- **The greater the difference in electronegativity between two atoms, the more polar their bond.**
- **Polar molecule:** A molecule in which the centers of positive and negative charge do not coincide.
- **Dipole:** Two electrical charges of equal magnitude but opposite sign are separated by a distance.
- **Dipole moment:** The quantitative measure of the magnitude of a dipole.

$$\mu = Qr$$
 (unit: Debyes, $1D = 3.34 \times 10^{-30} \text{ C} \cdot \text{m}$)
- **Differentiating ionic and covalent bonding:**
 - 1) Simplest approach: the interaction between a metal and a nonmetal is ionic and that between two nonmetals is covalent;
 - 2) More sophisticated approach: to use the difference in electronegativity as the main criterion for determining whether ionic or covalent bonding will be dominant.
 - 3) Properties of compounds are often best: Lower melting points mean covalent bonding, for example.
- **The procedures follow in the drawing Lewis structures:**
 - 1) Sum the valence electrons from all atoms, taking into account overall charge;
 - 2) Write the symbols for the atoms, show which atoms are attached to which, and connect them with a single bond (a line, representing two electrons);
 - 3) Complete the octets around all the atoms bonded to the central atom;
 - 4) Place any leftover electrons on the central atom;
 - 5) If there are not enough electrons to give the central atom an octet, try multiple bonds;
- **Formal charge:** the charge the atom would have if each bonding electron pair in the molecule were shared equally between its two atoms, different from the real charges on atoms.
- **To calculate the formal charge:** calculated by subtracting the number of electrons assigned to the atom from the number of valence electrons in the neutral atom.
 - 1) All unshared (nonbonding) electrons are assigned to the atom on which they are found;
 - 2) For any bond—single, double, or triple—**half of** the bonding electrons are assigned to each atom in the bond;

$$\text{Formal charge} = \text{valence electrons} - \frac{1}{2}(\text{bonding electrons}) - \text{nonbonding electrons}$$

- **Dominant Lewis structure:** the most important Lewis structure.
- **To choose the dominant Lewis structure:**
 - 1) The dominant Lewis structure is generally the one in which the atoms bear formal charges closest to zero;
 - 2) A Lewis structure in which any negative charges reside on the more electronegative atoms is generally more dominant than one that has negative charges on the less electronegative atoms;
- **Resonance:** a way of describing delocalized electrons within certain molecules or polyatomic ions where the bonding cannot be expressed by one single Lewis

structure. A molecule or ion with such delocalized electrons is represented by resonance structures.

- **Exceptions to the octet rule:**

- 1) Molecules and polyatomic ions containing an odd number of electrons;
 - 2) Molecules and polyatomic ions in which an atom has fewer than an octet of valence electrons;
 - 3) Molecules and polyatomic ions in which an atom has more than an octet of valence electrons.
- **Hypervalent:** molecules and ions with more than an octet of electrons around the central atom.

- **The strength of a covalent bond between two atoms is determined by the energy required to break the bond.**

- **Bond enthalpy:** the enthalpy change ΔH , for the breaking of a particular bond in one mole of a gaseous substance.

- **Atomization:** a molecule decomposed into its atoms.

- **Rules about bond enthalpy:**

- 1) Energy is always required to break chemical bonds.
- 2) Energy is always released when a bond forms between two gaseous atoms or molecular fragments.
- 3) **The greater the bond enthalpy, the stronger the bond is.**
- 4) A molecule with strong chemical bonds generally has less tendency to undergo chemical change than does one with weak bonds.

- $\Delta H_{rxn} = \Sigma(\text{bond enthalpies of bond broken}) - \Sigma(\text{bond enthalpies of bonds formed})$